Group 9 - Activity 6 Progress Report

BATUMBAKAL GYM MEMBERSHIP MANAGEMENT SYSTEM

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Database Backup Restoration and Recovery

Repository: https://github.com/louisandreidg/Batumbakal-Gym

Tracker: https://docs.google.com/spreadsheets/d/1BClCa9rEs9o-D9Rp3tUZVhECipJ234zX/edit?gid=2123136180

Evidence: https://github.com/louisandreidg/Batumbakal-Gym/tree/main/db-backup-recovery

Summary

We backed up each local project database, restored it into separate test databases and checked the restored objects, data and permissions. Recovery was tested by deleting one labeled record from a disposable copy and restoring it from a verified checkpoint backup.

Project	Restore result	Short notes
G01 DropBoxed	**Success with issues**	Backup and recovery worked; the existing direct-grant runtime design still needs review.
G02 Eventer / School Event Management	**Success**	Backup, restore, access checks and recovery worked.
G03 PAWPAW Veterinary	**Success with issues**	Backup and recovery worked; the existing direct-grant runtime design still needs review.
G04 HealthBridge Clinic	**Success with issues**	Backup and recovery worked after mapping owner-specific default privileges to the test owner.
G05 SHIORI Cafe	**Success**	Backup, restore, access checks and recovery worked.
G06 ByaheRentals / Car Rental	**Success with issues**	Backup and recovery worked after mapping owner-specific default privileges to the test owner.
G07 Applicant Management	**Success with issues**	Backup and recovery worked after mapping owner-specific default privileges to the test owner.
G08 Catering Management	**Success**	Backup, restore, access checks and recovery worked.

Project	Restore result	Short notes
G10 UniSync	**Success**	Backup, restore, access checks and recovery worked.
G11 ReserveIT	**Success**	Backup, restore, access checks and recovery worked.

All recovery tests used separate databases. Original application data and the initial archives were preserved. Backup files are stored locally with restricted access and are not included in GitHub.

Backup and recovery plan

The backups use pg_dump custom format and are inspected with pg_restore --list. Each archive has a recorded file size, SHA-256 checksum and creation time. A successful restore and data comparison, rather than the file existing alone, provide the recovery evidence.

Dedicated backup accounts have SELECT access to each source database. Separate restore accounts own the disposable databases and have no superuser, role-creation or database-creation privilege. The verifier inherits the existing Activity 4 runtime privileges and is restricted to the test databases. No existing application password was reset.

Archives are held in the protected local Activity6-Private folder. Only the current Windows user and SYSTEM have filesystem access. The archives are compressed, not encrypted; a real deployment should also encrypt them and keep an off-device copy.

Proposed retention is the last day of frequent copies, seven daily copies, four weekly copies and three monthly copies. Restore tests should run monthly and after schema changes. The following are proposed deployment objectives, not service guarantees.

Project	Backup interval and RPO	Proposed RTO
G01 DropBoxed	24 hours	4 hours
G02 Eventer / School Event Management	4 hours	2 hours
G03 PAWPAW Veterinary	1 hour	1 hour

Project	Backup interval and RPO	Proposed RTO
G04 HealthBridge Clinic	1 hour	1 hour
G05 SHIORI Cafe	1 hour	1 hour
G06 ByaheRentals / Car Rental	1 hour	2 hours
G07 Applicant Management	24 hours	4 hours
G08 Catering Management	1 hour	2 hours
G10 UniSync	4 hours	2 hours
G11 ReserveIT	1 hour	1 hour

Hourly copies suit the booking and transaction-oriented projects; the longer intervals assume the group can accept a larger loss window. The owning group should confirm those assumptions. Backups cannot recover later changes that they never captured; shorter loss windows need more frequent copies or a tested WAL-based recovery plan.

G01 DropBoxed

Repository: https://github.com/joshxaaaaaa/GRP1_Dropboxed

Database: dropboxed_db

Result: Success with issues. Backup and recovery worked; the existing direct-grant runtime design still needs review.

Backup and restore

Backup account: a6_g01_backup (a6_g01_backup_grp). Restore owner: a6_g01_restore. Archive: g01_source_run1.dump; 37,023 bytes; created 07 November 2026 at 01:47:08 UTC+8.

SHA-256: a2ca8d57473f394263c2208dce1387e5ee686d2c8bbe1bc1ff77acdde1cb3af8

Command: pg_dump -w -h localhost -U a6_g01_backup -Fc --verbose -d dropboxed_db > g01_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g01_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g01_restore_20261107 g01_source_run1.dump

Check	Result
Tables and data	8 tables; 5 rows. Counts and aggregate row fingerprints matched. dropboxed.complaint_categories: 4 rows.
Keys and objects	8 primary keys, 8 foreign keys, 72 total constraints, 22 indexes and 8 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g01_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G01-20261107 was created only in a6_g01_restore_20261107. The separate checkpoint g01_checkpoint_run2.dump was restored and checked in a6_g01_preflight_20261107 before the incident.

At 07 November 2026 at 01:48:57 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g01_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.192 seconds; restore plus verification: 0.715 seconds.

Notes

The existing runtime account still uses direct grants. Its privileges were inherited by the clone verifier for this test; the Activity 4 privilege-group design still needs separate work.

Original Node.js connection module connected successfully to the restored copy. This was a connection check, not a full web workflow test.

Proposed RPO: 24 hours; RTO: 4 hours. The measured local restore time does not include a full application restart.

G02 Eventer / School Event Management

Repository: https://github.com/JosephEmmanuelBabas/School_Event_Management_Web_Appli

Database: grp02_event_management_db

Result: Success. Backup, restore, access checks and recovery worked.

Backup and restore

Backup account: a6_g02_backup (a6_g02_backup_grp). Restore owner: a6_g02_restore. Archive: g02_source_run1.dump; 41,905 bytes; created 07 November 2026 at 01:49:08 UTC+8.

SHA-256: 95bc72000eaaa89efe88c5b177a475d5eac58d9bd69dab63ebe17ae1ea284293

Command: pg_dump -w -h localhost -U a6_g02_backup -Fc --verbose -d grp02_event_management_db > g02_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g02_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g02_restore_20261107 g02_source_run1.dump

Check	Result
Tables and data	10 tables; 92 rows. Counts and aggregate row fingerprints matched. school_event_management_db.audit_log: 84 rows.
Keys and objects	10 primary keys, 8 foreign keys, 83 total constraints, 27 indexes and 9 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g02_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G02-20261107 was created only in a6_g02_restore_20261107. The separate checkpoint g02_checkpoint_run1.dump was restored and checked in a6_g02_preflight_20261107 before the incident.

At 07 November 2026 at 01:49:10 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g02_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.274 seconds; restore plus verification: 0.929 seconds.

Notes

Object ownership was mapped to the isolated restore owner. Original runtime and reporting grants were retained.

The runtime-role SQL connection and query were tested. The web application was not reconfigured for this copy.

Proposed RPO: 4 hours; RTO: 2 hours. The measured local restore time does not include a full application restart.

G03 PAWPAW Veterinary

Repository: https://github.com/rubiakylashane/GRP03_PAWPAW

Database: group3_Pawpaw

Result: Success with issues. Backup and recovery worked; the existing direct-grant runtime design still needs review.

Backup and restore

Backup account: a6_g03_backup (a6_g03_backup_grp). Restore owner: a6_g03_restore. Archive: g03_source_run1.dump; 46,742 bytes; created 07 November 2026 at 01:49:12 UTC+8.

SHA-256: a291ab33eafc17ead8b85c24ad5032071b899b5263ca845fc0476a4fd8176f10

Command: pg_dump -w -h localhost -U a6_g03_backup -Fc --verbose -d group3_Pawpaw > g03_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g03_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g03_restore_20261107 g03_source_run1.dump

Check	Result
Tables and data	14 tables; 9 rows. Counts and aggregate row fingerprints matched. public.service: 9 rows.
Keys and objects	14 primary keys, 16 foreign keys, 118 total constraints, 23 indexes and 12 sequences matched.
Views and functions	0 views and 1 user functions matched; extension versions matched.
Access control	a6_g03_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G03-20261107 was created only in a6_g03_restore_20261107. The separate checkpoint g03_checkpoint_run2.dump was restored and checked in a6_g03_preflight_20261107 before the incident.

At 07 November 2026 at 01:51:49 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g03_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.191 seconds; restore plus verification: 0.73 seconds.

Notes

The existing runtime account still uses direct grants. Its privileges were inherited by the clone verifier for this test; the Activity 4 privilege-group design still needs separate work.

Original Node.js connection module connected successfully to the restored copy. This was a connection check, not a full web workflow test.

Proposed RPO: 1 hour; RTO: 1 hour. The measured local restore time does not include a full application restart.

G04 HealthBridge Clinic

Repository: https://github.com/RainneQuintab/GRP03_PROJECTDB

Database: grp03_projectdb

Result: Success with issues. Backup and recovery worked after mapping owner-specific default privileges to the test owner.

Backup and restore

Backup account: a6_g04_backup (a6_g04_backup_grp). Restore owner: a6_g04_restore. Archive: g04_source_run1.dump; 49,475 bytes; created 07 November 2026 at 01:49:13 UTC+8.

SHA-256: f5bcb016f7140e1dd36286fd7689805786a078c4083f3a6d053a5602084f62b7

Command: pg_dump -w -h localhost -U a6_g04_backup -Fc --verbose -d grp03_projectdb > g04_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g04_restore --verbose --exit-on-error --single-transaction --no-owner --use-list run2-initial-restore-restore-list.txt -d a6_g04_restore_20261107 g04_source_run1.dump

Check	Result
Tables and data	11 tables; 21 rows. Counts and aggregate row fingerprints matched. public.appointments: 2 rows.
Keys and objects	11 primary keys, 9 foreign keys, 90 total constraints, 22 indexes and 11 sequences matched.
Views and functions	2 views and 0 user functions matched; extension versions matched.
Access control	a6_g04_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G04-20261107 was created only in a6_g04_restore_20261107. The separate checkpoint g04_checkpoint_run2.dump was restored and checked in a6_g04_preflight_20261107 before the incident.

At 07 November 2026 at 01:51:56 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g04_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.209 seconds; restore plus verification: 1.032 seconds.

Notes

The first restore stopped on default privileges owned by the original owner. Those archive entries were explicitly excluded from the restore list, then replayed for the isolated owner. The mapped default grants were compared with the source and matched. Ordinary object ACLs were retained.

The runtime-role SQL connection and query were tested. The web application was not reconfigured for this copy.

Proposed RPO: 1 hour; RTO: 1 hour. The measured local restore time does not include a full application restart.

G05 SHIORI Cafe

Repository: https://github.com/LeeJnsn27/SHIORIcafe-new

Database: grp10_shioricafedb

Result: Success. Backup, restore, access checks and recovery worked.

Backup and restore

Backup account: a6_g05_backup (a6_g05_backup_grp). Restore owner: a6_g05_restore. Archive: g05_source_run1.dump; 89,339 bytes; created 07 November 2026 at 01:49:14 UTC+8.

SHA-256: cd73d0b27d739368c43efc6f163b9d7431caba1e609ffca50284c3ed861b900b

Command: pg_dump -w -h localhost -U a6_g05_backup -Fc --verbose -d grp10_shioricafedb > g05_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g05_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g05_restore_20261107 g05_source_run1.dump

Check	Result
Tables and data	17 tables; 23 rows. Counts and aggregate row fingerprints matched. shioricafe_schema.category: 2 rows.
Keys and objects	17 primary keys, 24 foreign keys, 217 total constraints, 49 indexes and 17 sequences matched.
Views and functions	3 views and 3 user functions matched; extension versions matched.
Access control	a6_g05_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G05-20261107 was created only in a6_g05_restore_20261107. The separate checkpoint g05_checkpoint_run1.dump was restored and checked in a6_g05_preflight_20261107 before the incident.

At 07 November 2026 at 01:49:17 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g05_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.351 seconds; restore plus verification: 0.98 seconds.

Notes

Object ownership was mapped to the isolated restore owner. Original runtime and reporting grants were retained.

Original SHIORI db.php with separate private test configuration connected successfully to the restored copy. This was a connection check, not a full web workflow test.

Proposed RPO: 1 hour; RTO: 1 hour. The measured local restore time does not include a full application restart.

G06 ByaheRentals / Car Rental

Repository: https://github.com/123airabels/Car_Rental_Management_System

Database: car_rental_db

Result: Success with issues. Backup and recovery worked after mapping owner-specific default privileges to the test owner.

Backup and restore

Backup account: a6_g06_backup (a6_g06_backup_grp). Restore owner: a6_g06_restore. Archive: g06_source_run1.dump; 81,697 bytes; created 07 November 2026 at 01:49:18 UTC+8.

SHA-256: 93a6621652610f5728c660742191da60df06ddaa9603e11881a0ad2103761f64

Command: pg_dump -w -h localhost -U a6_g06_backup -Fc --verbose -d car_rental_db > g06_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g06_restore --verbose --exit-on-error --single-transaction --no-owner --use-list run2-initial-restore-restore-list.txt -d a6_g06_restore_20261107 g06_source_run1.dump

Check	Result
Tables and data	17 tables; 26 rows. Counts and aggregate row fingerprints matched. public.audit_log: 3 rows.
Keys and objects	17 primary keys, 25 foreign keys, 171 total constraints, 38 indexes and 17 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g06_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G06-20261107 was created only in a6_g06_restore_20261107. The separate checkpoint g06_checkpoint_run2.dump was restored and checked in a6_g06_preflight_20261107 before the incident.

At 07 November 2026 at 01:52:00 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g06_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.221 seconds; restore plus verification: 0.811 seconds.

Notes

The first restore stopped on default privileges owned by the original owner. Those archive entries were explicitly excluded from the restore list, then replayed for the isolated owner. The mapped default grants were compared with the source and matched. Ordinary object ACLs were retained.

The runtime-role SQL connection and query were tested. The web application was not reconfigured for this copy.

Proposed RPO: 1 hour; RTO: 2 hours. The measured local restore time does not include a full application restart.

G07 Applicant Management

Repository: https://github.com/mang2mas/grp07-applicant-management-system

Database: grp07_applicantdb

Result: Success with issues. Backup and recovery worked after mapping owner-specific default privileges to the test owner.

Backup and restore

Backup account: a6_g07_backup (a6_g07_backup_grp). Restore owner: a6_g07_restore. Archive: g07_source_run1.dump; 36,062 bytes; created 07 November 2026 at 01:49:20 UTC+8.

SHA-256: 05fe1275647799460bba767e93abc8b2f18ff2026d54b965b26e46b3cbe1dd93

Command: pg_dump -w -h localhost -U a6_g07_backup -Fc --verbose -d grp07_applicantdb > g07_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g07_restore --verbose --exit-on-error --single-transaction --no-owner --use-list run2-initial-restore-restore-list.txt -d a6_g07_restore_20261107 g07_source_run1.dump

Check	Result
Tables and data	8 tables; 74 rows. Counts and aggregate row fingerprints matched. public.applicant: 10 rows.
Keys and objects	8 primary keys, 7 foreign keys, 63 total constraints, 20 indexes and 7 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g07_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G07-20261107 was created only in a6_g07_restore_20261107. The separate checkpoint g07_checkpoint_run3.dump was restored and checked in a6_g07_preflight_20261107 before the incident.

At 07 November 2026 at 01:52:32 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g07_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.162 seconds; restore plus verification: 0.731 seconds.

Notes

The first restore stopped on default privileges owned by the original owner. Those archive entries were explicitly excluded from the restore list, then replayed for the isolated owner. The mapped default grants were compared with the source and matched. Ordinary object ACLs were retained.

Original PHP database.php db() function connected successfully to the restored copy. This was a connection check, not a full web workflow test.

Proposed RPO: 24 hours; RTO: 4 hours. The measured local restore time does not include a full application restart.

G08 Catering Management

Repository: https://github.com/yami-sumi/catering-management-system

Database: catering_management_db

Result: Success. Backup, restore, access checks and recovery worked.

Backup and restore

Backup account: a6_g08_backup (a6_g08_backup_grp). Restore owner: a6_g08_restore. Archive: g08_source_run1.dump; 36,513 bytes; created 07 November 2026 at 01:49:21 UTC+8.

SHA-256: b06a7366da87f2fcdb581e3cf87f8b24350eacbd9a558c50a0429742e19ae6ef

Command: pg_dump -w -h localhost -U a6_g08_backup -Fc --verbose -d catering_management_db > g08_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g08_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g08_restore_20261107 g08_source_run1.dump

Check	Result
Tables and data	12 tables; 35 rows. Counts and aggregate row fingerprints matched. public.admin: 1 rows.
Keys and objects	12 primary keys, 8 foreign keys, 81 total constraints, 16 indexes and 9 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g08_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G08-20261107 was created only in a6_g08_restore_20261107. The separate checkpoint g08_checkpoint_run1.dump was restored and checked in a6_g08_preflight_20261107 before the incident.

At 07 November 2026 at 01:49:23 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g08_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.446 seconds; restore plus verification: 0.944 seconds.

Notes

Object ownership was mapped to the isolated restore owner. Original runtime and reporting grants were retained.

Original Node.js connection module connected successfully to the restored copy. This was a connection check, not a full web workflow test.

Proposed RPO: 1 hour; RTO: 2 hours. The measured local restore time does not include a full application restart.

G10 UniSync

Repository: https://github.com/CoffyJosh/unisync

Database: grp10_unisyncdb

Result: Success. Backup, restore, access checks and recovery worked.

Backup and restore

Backup account: a6_g10_backup (a6_g10_backup_grp). Restore owner: a6_g10_restore. Archive: g10_source_run1.dump; 66,277 bytes; created 07 November 2026 at 01:49:24 UTC+8.

SHA-256: 8742812c5e3d76523c9d1ee2d5eb2231b84e90a16523c072ac33db8ef9f595f2

Command: pg_dump -w -h localhost -U a6_g10_backup -Fc --verbose -d grp10_unisyncdb > g10_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g10_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g10_restore_20261107 g10_source_run1.dump

Check	Result
Tables and data	8 tables; 1573 rows. Counts and aggregate row fingerprints matched. public.account: 130 rows.
Keys and objects	8 primary keys, 11 foreign keys, 80 total constraints, 21 indexes and 8 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g10_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G10-20261107 was created only in a6_g10_restore_20261107. The separate checkpoint g10_checkpoint_run1.dump was restored and checked in a6_g10_preflight_20261107 before the incident.

At 07 November 2026 at 01:49:26 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g10_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.276 seconds; restore plus verification: 0.913 seconds.

Notes

Object ownership was mapped to the isolated restore owner. Original runtime and reporting grants were retained.

The runtime-role SQL connection and query were tested. The web application was not reconfigured for this copy.

Proposed RPO: 4 hours; RTO: 2 hours. The measured local restore time does not include a full application restart.

G11 ReserveIT

Repository:
https://github.com/sev-villanueva/ReserveIT-Classroom-and-Material-Reservation-System-

Database: ReserveIT

Result: Success. Backup, restore, access checks and recovery worked.

Backup and restore

Backup account: a6_g11_backup (a6_g11_backup_grp). Restore owner: a6_g11_restore. Archive: g11_source_run1.dump; 41,311 bytes; created 07 November 2026 at 01:49:27 UTC+8.

SHA-256: 2361aee2673097130edcf6e88fa474edf0b348ec5ed4301b92ec5fbfd3bd471d

Command: pg_dump -w -h localhost -U a6_g11_backup -Fc --verbose -d ReserveIT > g11_source_run1.dump

Restore: pg_restore -w -h localhost -U a6_g11_restore --verbose --exit-on-error --single-transaction --no-owner -d a6_g11_restore_20261107 g11_source_run1.dump

Check	Result
Tables and data	8 tables; 45 rows. Counts and aggregate row fingerprints matched. public.app_user: 4 rows.
Keys and objects	8 primary keys, 7 foreign keys, 86 total constraints, 19 indexes and 7 sequences matched.
Views and functions	0 views and 0 user functions matched; extension versions matched.
Access control	a6_g11_verify ran the application SELECT successfully; schema CREATE was denied. Existing grants and memberships matched.
Representative SELECT	The source and restored query results had matching row counts and aggregate fingerprints.

Recovery test

A probe row with ID 1 and marker COMP018-A6-G11-20261107 was created only in a6_g11_restore_20261107. The separate checkpoint g11_checkpoint_run2.dump was restored and checked in a6_g11_preflight_20261107 before the incident.

At 07 November 2026 at 01:51:52 UTC+8, the probe was deleted from the incident copy and its count became zero. The checkpoint was then restored into a6_g11_recovery_20261107. The probe returned with the same marker, and the application data still matched. Recovery result: Recovered. Restore command time: 0.269 seconds; restore plus verification: 0.801 seconds.

Notes

Object ownership was mapped to the isolated restore owner. Original runtime and reporting grants were retained.

The runtime-role SQL connection and query were tested. The web application was not reconfigured for this copy.

Proposed RPO: 1 hour; RTO: 1 hour. The measured local restore time does not include a full application restart.

Recovery evaluation

Each source archive was inspected and restored successfully. The checkpoint was also restored before deletion, so the incident was simulated only after the needed record had a proven recovery copy. File checksums were checked before restores and the initial archive was checked again afterward.

The fixture was added after the initial source backup. It could be recovered from the later checkpoint, not from the earlier source archive. In a real incident, changes after the latest backup would be lost unless another backup or WAL recovery covered them.

Restored owners differ from the source owners by design. Roles are cluster-wide and are not supplied by a single-database dump; the current cluster already had the Activity 4 roles. Password-free reconstruction scripts are included for a fresh environment, where passwords must be set privately. A separate fresh-cluster deployment was not tested.

Some CHECK expressions were deparsed with equivalent text-array casts. The comparison normalized only literal ANY/ALL array forms and retained the surrounding logic. Original definitions are saved for review. Owner-specific default privileges were replayed for the test owner, with original errors and explicit restore lists retained.

The original data was unchanged during the checks. These were small local test databases; the timings are not production recovery guarantees. Full web workflows, external files, uploads, email

services and payment services were not restored or tested. Application files and server configuration need their own recovery plan.

Recommendations and feedback

Keep encrypted off-device backups and test the restore process regularly. Document role creation, ownership and future grants alongside the application schema. Review the proposed RPO/RTO with the owners and test the complete application before accepting a production recovery plan. Dropboxed and PAWPAW still need their earlier runtime privilege-group work.

Feedback drafts are saved in each project folder. No Activity 6 feedback has been sent. The class tracker still needs the Activity 6 entries.

AI usage declaration

AI was used to help us prepare and run the backup and restore commands. Commands were checked against PostgreSQL documentation and the saved output. Recovery was verified using the restored data and the disposable test record.